Gauri Toshniwal

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EDUCATION

National Institute of Technology, Tiruchirappalli — Tamil Nadu, India
*Masters of Technology - Data Analytics (CGPA: **9.58**/10) **[Rank 1]*** — 08/2023 - 06/2025
Awarded Reliance Foundation Postgraduate Scholarship for the year 2023-24
Coursework: Deep Learning, **NLP**, **Machine Learning**, Federated Learning, Mathematics (Linear Algebra, Probability)

Government College of Engineering, Nagpur — Maharashtra, India
*Bachelor of Engineering - Computer Science and Engineering (CGPA: **9.42**/10)* — 08/2017 - 06/2021
Coursework: Operating Systems, Data Structures & Algorithms, Artificial Intelligence, Databases, Discrete Mathematics, Compilers

RESEARCH

- **Contextual QA for Financial Causality Detection Combining Extractive and Generative Models** — 12/2024
  FinCausal 2025 @ COLING - 3rd Rank (English Track) **(Paper)**
- Designed an extractive generative hybrid system using RoBERTa-base for causal span detection and Gemma-2 9B for contextual justification generation.
- Achieved **SAS 96.74%** and **EM 70.14%**, improving causal reasoning consistency in financial narratives.

- **Object Detection in Indian Food Platters using Transfer Learning with YOLOv4** — 05/2022
  DECOR 2022 @ ICDE - Best Student Paper Award **(Paper)**
- Developed a domain-adapted YOLOv4 model and curated IndianFood10, a custom dataset for Indian cuisine recognition.

- **Do Easy Math Problems Help Models Learn Hard Ones?** — 07/2025
- Investigated difficulty and transfer in LLM mathematical reasoning using MATH and Llama-3.2-1B, Built a full pipeline with custom instruction formatting, label masking, LoRA (QLoRA 4-bit)
- Analyzed gradient-subspace geometry to characterize task transfer, showing that transfer follows model-level gradient proximity rather than human-defined difficulty.

WORK EXPERIENCE

Astreya — Remote, India
AI/ML Engineer (Google Extended Workforce) — 07/2025 - Current

- **CTI Verifier: High-Precision Ticket Classification & Preloading System**
- Researched and developed a production-ready **c**onfidence-aware selective classifier with an abstention mechanism for **275-class** hierarchical ticket classification, systematically evaluating retrieval, embedding, and LLM-based approaches.
- Built a confidence-calibrated classifier achieving **95.6% precision** on accepted predictions, enabling reliable automation of **~55K** manual ticket classification decisions annually.
- **Corvus: Enterprise AI Copilot for IT Operations**
- Built and productized an enterprise AI copilot, enabling SOP-grounded chat, next-step recommendations, quality audits, and context-aware AI workflows with Redis-backed state and BigQuery telemetry.
- **Cut playbook search and ticket resolution time**, delivering **~10 FTE annual savings** through AI-powered automation, grounded recommendations, and quality-assurance workflows.

Findability Sciences — Remote, India
Data Scientist (Onsite - Hindustan Times Digital) — 05/2021 - 08/2023

- **Demographics (Gender, Age, Interest) Prediction:**
- Built user-level classifiers for gender, age, and interest prediction using grammatical and entity-based features from news headlines, achieving an F1 score of 0.65 for **2M** users daily. Eliminated reliance on third-party data, saving **₹500,000/month**.
- **Subscription Churn Prediction and Retention Feed:**
- Developed XGBoost models (d-60, d-30, d-15) to track churn probability shifts and used **SHAP** for interpretability powering a personalized **retention feed** that improved user engagement.

CERTIFICATION

- **ACM Summer School on Responsible and Safe AI Certificate** — 06/2024
- **AI Safety Fundamentals - Bluedot Impact Certificate** — 03/2024 - 06/2024

SKILLS

Languages: Python, C, SQL
ML & AI: PyTorch, Scikit-learn, Pandas, NumPy, PySpark, Hugging Face Transformers, RAG, LLM Evaluation
Frameworks, Cloud & Tools: FastAPI, Flask, Git, Docker, GCP, Vertex AI, BigQuery ML, MongoDB, Redis
Areas of Interest: NLP, LLMs, Deep Learning, AI Evaluation, AI Safety